Inferential Statistic Project: Statistic Model for Neonatal Weight Forecasting

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2025-02-12

Preliminary Analysis

Dataset presentation

The dataset presents 2500 observations, for 10 variables:
* Anni.madre: continuous quantitative variable ;
* N.gravidanze: discrete quantitative variable ;
* Fumatrice: Factor/dummy variable: “0” means non-smoker while “1” means smoker ;
* Gestazione: continuous quantitative variable ;
* Peso: continuous quantitative variable ;
* Lunghezza: continuous quantitative variable ;
* Cranio: continuous quantitative variable ;
* Tipo.parto: qualitative variable on nominal scale with 2 modalities: “Ces” and “Nat” ;
* Ospedale: qualitative variable on nominal scale with 3 modalities: “osp1”. “osp2” and “osp3” ;
* Sesso: qualitative variable on nominal scale with 2 modalities: “F” and “M”.

In this study, “Peso” is the response variable, while the others are predictor variables.
In particular “Lunghezza” and “Cranio” are anthropometric data, used as control variables.

Description of quantitative variables

```
data <- read.csv("neonati.csv", sep=",", stringsAsFactors = T)
n<-dim(data)[1]
attach(data)

matrix_data <- matrix(nrow = 6, ncol = 6, byrow=F)

for (j in 1:6) {
  matrix_data[,j] <- c(summary(Anni.madre)[j], summary(N.gravidanze)[j], summary(Gestazione)[j], summary(Peso)[j], summary(L
unghezza)[j], summary(Cranio)[j])
  }

library(DT)
datatable(round(matrix_data, digits =2),
  colnames = c("Min", "1st. Quart", "Median", "Mean", "3rd Quart", "Max"),
  rownames = c("Anni.madre", "N.gravidanze", "Gestazione", "Peso", "Lunghezza", "Cranio")
)
```

Show 10 entries Search:

	Min	1st. Quart	Median	Mean	3rd Quart	Max
Anni.madre	0	25	28	28.16	32	46
N.gravidanze	0	0	1	0.98	1	12
Gestazione	25	38	39	38.98	40	43
Peso	830	2990	3300	3284.08	3620	4930
Lunghezza	310	480	500	494.69	510	565
Cranio	235	330	340	340.03	350	390

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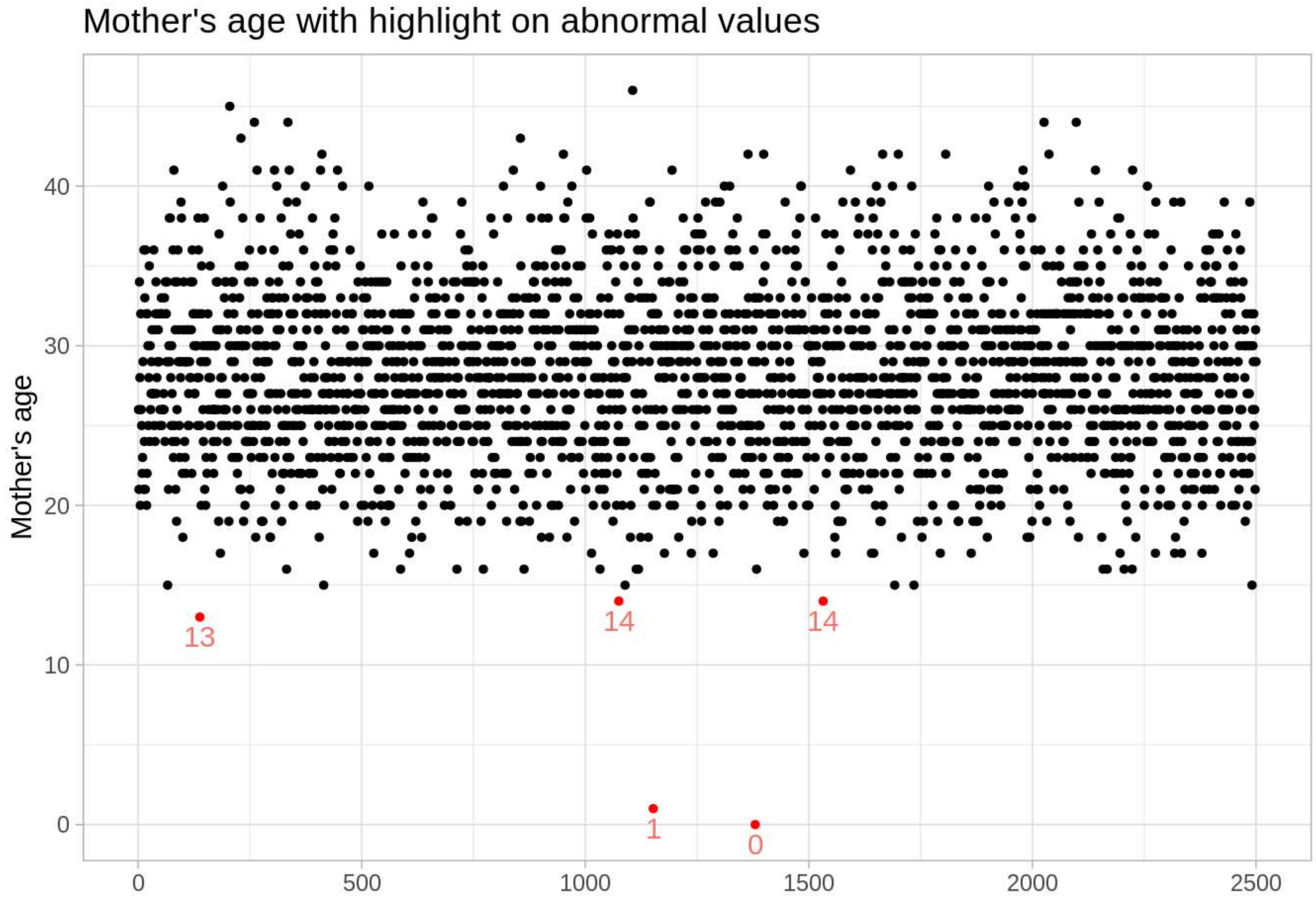
The first variable, mother’s age, presents a minimum at 0 years old. To deeply investigate, a plot is done, in order to highlight eventual outliers/anomalies on low mother’s age value.

```
library(ggplot2)
ggplot(data,aes(x=1:n,y=Anni.madre,label=ifelse(Anni.madre<(quantile(Anni.madre, 0.25) - 1.5 * IQR(Anni.madre)), Anni.madre
,"")))+
  geom_point(size=1,color = ifelse(Anni.madre < (quantile(Anni.madre, 0.25) - 1.5 * IQR(Anni.madre)), "red", "black"))+
  geom_text(aes(colour="red"), vjust = 1.5, show.legend=F)+
  labs(title="Mother's age with highlight on abnormal values", x="", y="Mother's age")+
  theme_light()
```



It appears that among the 5 values identified as outliers, 2 are errors with ages of 0 and 1 year old.

These statistic units are removed from that database, as they can't correspond to true values.

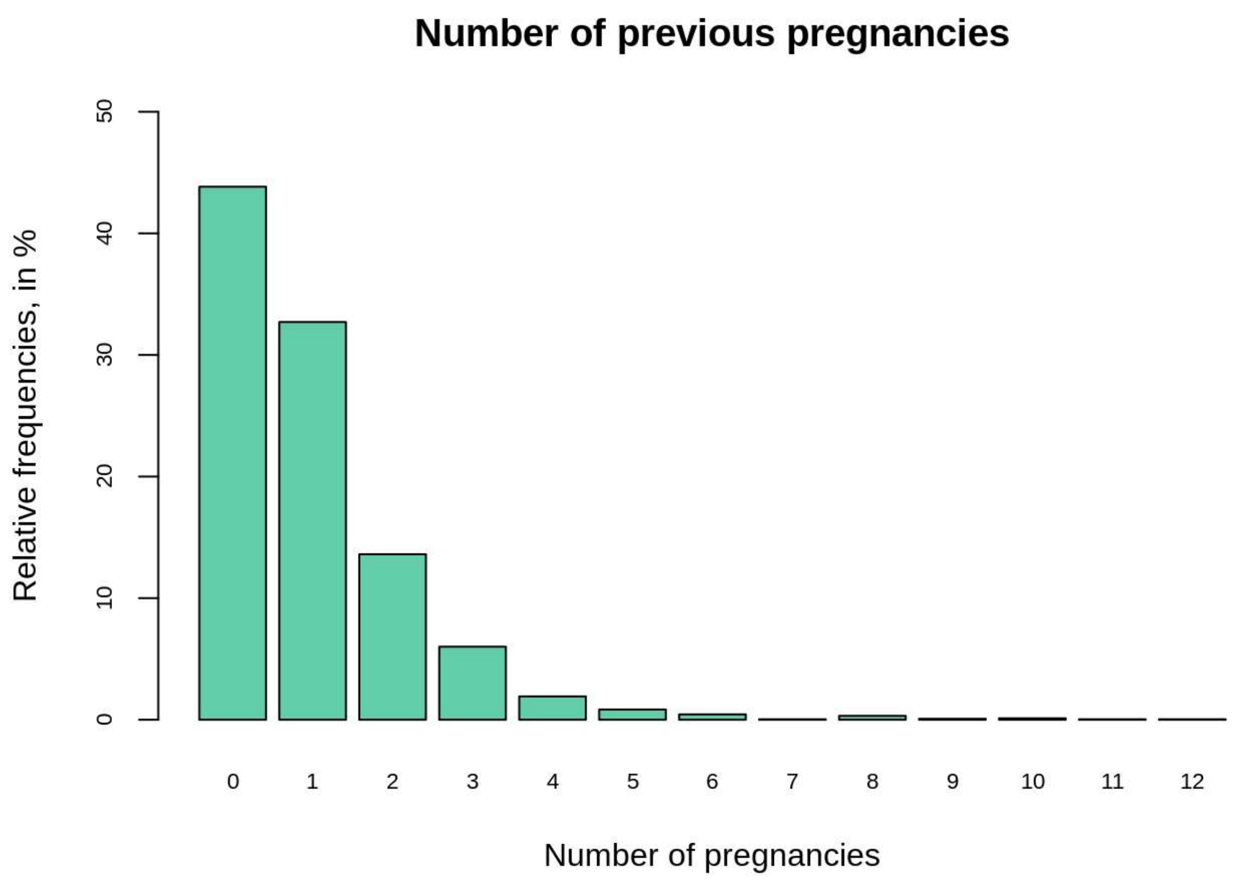
```
new_data <- data[ !(Anni.madre %in% c(0, 1)), ]
n<-dim(new_data)[1]
detach(data)
attach(new_data)
```

Going back to the recapitulative table of quantitative variables, there are no abnormal values in the min and max. To fully describe them, some plots are performed.

Number of previous pregnancies

```
fi <- table(N.gravidanze)/n*100

barplot(fi,
        main ="Number of previous pregnancies" ,
        xlab ="Number of pregnancies" ,
        ylab ="Relative frequencies, in %",
        ylim = c(0,50),
        cex.axis = 0.7,
        cex.names= 0.7,
        col= "mediumaquamarine")
```

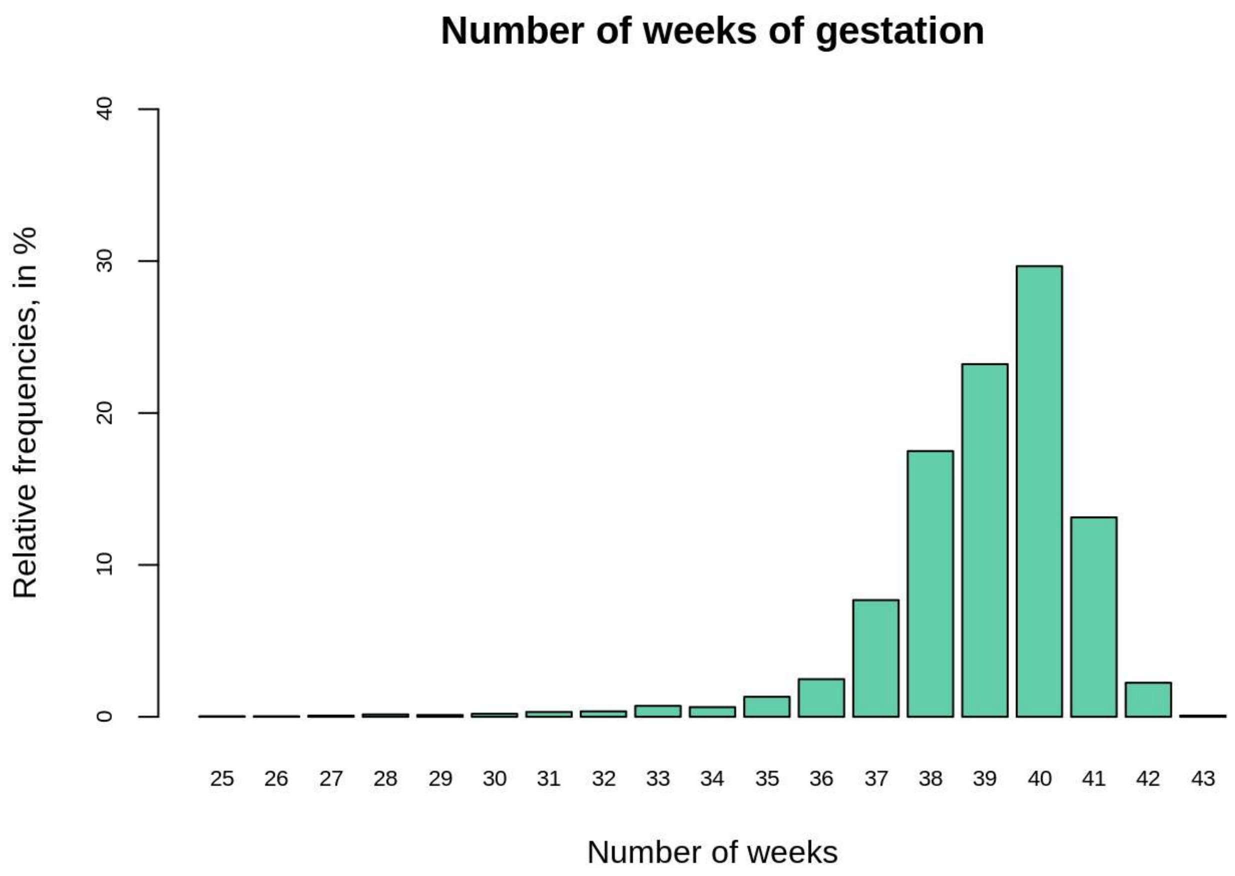


For more than 40% of the mothers, it is a first pregnancy. The mother with major previous pregnancies had 12.
The mean value is 0.98 pregnancy, very close value to 1. So, on average it is the second pregnancy.

Number of weeks of gestation

```
fi <- table(Gestazione)/n*100

barplot(fi,
        main ="Number of weeks of gestation" ,
        xlab ="Number of weeks" ,
        ylab ="Relative frequencies, in %",
        ylim = c(0,40),
        cex.axis = 0.7,
        cex.names= 0.7,
        col= "mediumaquamarine")
```



For nearly 30% of the births, it happens at the 40th week, which corresponds to the theoretical due date of the gestation. According to the summary table, the mean and median values are equal to 39 weeks. The minimum is 25 weeks, the maximum is 43 weeks. There is no abnormal value here.

Weight, length and cranium size

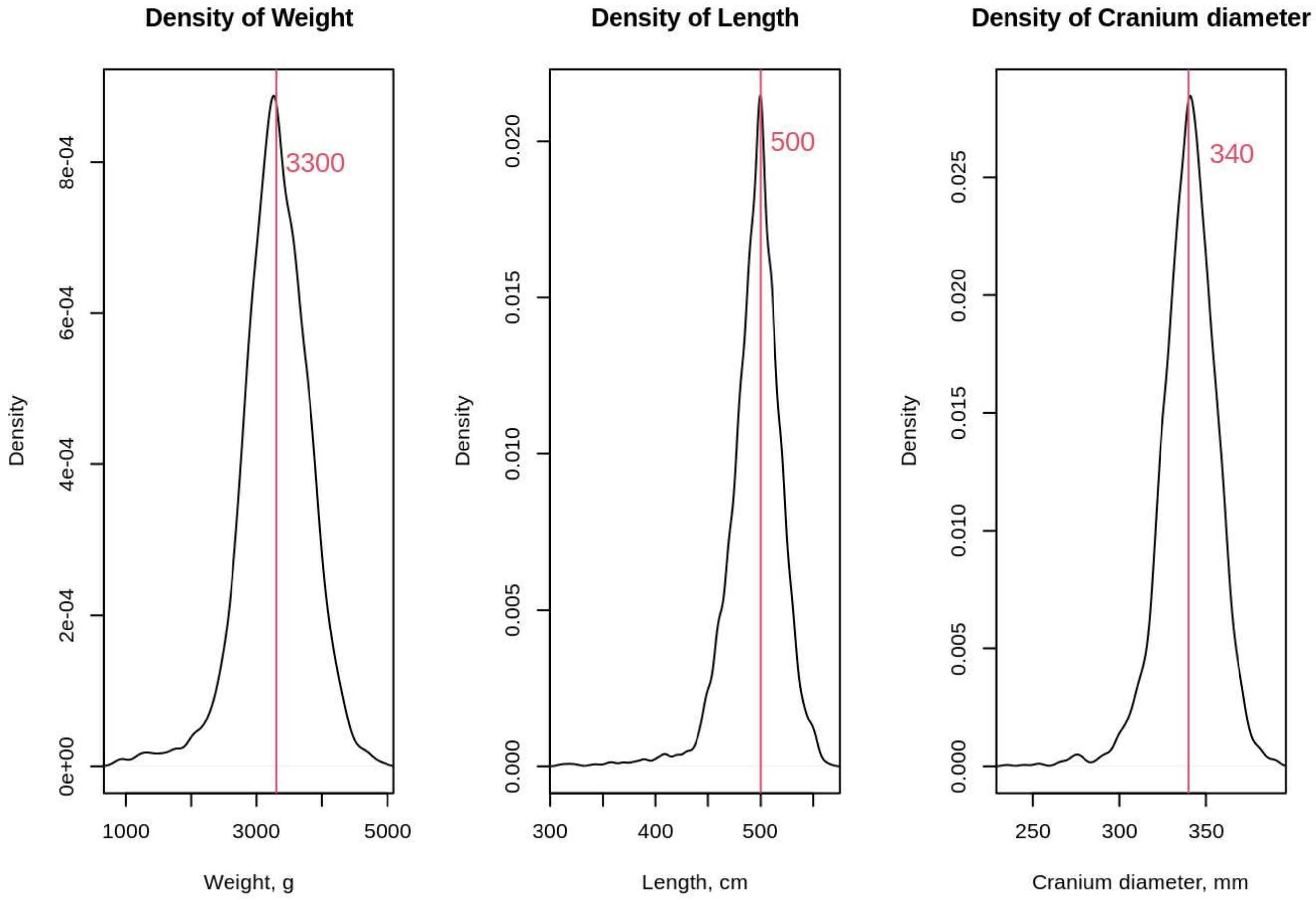
For these 3 continuous quantitative variables, we plot here their densities. As many variables found in nature, we expect Gaussian distributions.

```
par(mfrow=c(1,3))

plot(density(Peso),
     main = "Density of Weight",
     xlab = "Weight, g",
     ylab = "Density",
     xlim = c(min(Peso), max(Peso)))
abline(v=median(Peso), col=2)
text(x=median(Peso)+600, y=0.0008, labels = median(Peso), col=2, cex = 1.3)

plot(density(Lunghezza),
     main = "Density of Length",
     xlab = "Length, cm",
     ylab = "Density",
     xlim = c(min(Lunghezza), max(Lunghezza)))
abline(v=median(Lunghezza), col=2)
text(x=median(Lunghezza)+30, y=0.02, labels = median(Lunghezza), col=2, cex = 1.3)

plot(density(Cranio),
     main = "Density of Cranium diameter",
     xlab = "Cranium diameter, mm",
     ylab = "Density",
     xlim = c(min(Cranio), max(Cranio)))
abline(v=median(Cranio), col=2)
text(x=median(Cranio)+25, y=0.026, labels = median(Cranio), col=2, cex = 1.3)
```



Shape factors:

```
library(moments)
matrix_shape <- round(matrix(c
                       (skewness(Peso), skewness(Lunghezza), skewness(Cranio),
                         kurtosis(Peso)-3, kurtosis(Lunghezza)-3, kurtosis(Cranio)-3),
nrow = 3, ncol = 2, byrow=F), 2)

datatable(matrix_shape,
  colnames = c("Skewness", "Kurtosis"),
  rownames = c("Weight", "Length", "Cranium diameter"))
```

Show 10 entries Search:

	Skewness	Kurtosis
Weight	-0.65	2.03
Length	-1.51	6.48
Cranium diameter	-0.79	2.94

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Compared with Gaussian distributions, these 3 ones have negative skewnesses and are leptokurtic.

Description of qualitative variables

Smoking

```
fi_S <- round(table(Fumatrici)/n, 2)*100
print(fi_S)
```

```
## Fumatrici
##  0  1
## 96  4
```

In this survey, 96% of the mothers are non-smoking, while 4% smoke.

Type of delivery

```
fi_D <- round(table(Tipo.parto)/n, 2)*100
print(fi_D)
```

```
## Tipo.parto
## Ces Nat
##  29  71
```

71% of the mothers had a natural delivery, while 29% a C-section.

Hospital

```
fi_H <- round(table(Ospedale)/n, 2)*100
print(fi_H)
```

```
## Ospedale
## osp1 osp2 osp3
##   33   34   33
```

There is absolute equality: the survey considered the same number of mothers in each of the 3 hospitals.

Newborn sex

```
fi_Sex <- round(table(Sesso)/n, 2)*100
print(fi_Sex)
```

```
## Sesso
##  F  M
## 50 50
```

There is equality between girls and boys.

Hypothesis to test

Cesarean delivery is performed more often in some hospital

We want to know if the 2 qualitative variables “Ospedale” and “Tipo.parto” are independent. A Chi-squared test is performed.

```
table_Osp_Par <- table(Ospedale, Tipo.parto)
test_Osp_Par <- chisq.test(table_Osp_Par)
print(test_Osp_Par)
```

```
##
##  Pearson's Chi-squared test
##
## data:  table_Osp_Par
## X-squared = 1.083, df = 2, p-value = 0.5819
```

p-value is equal to 0.58, much higher than 5%, so we can’t refuse H0 hypothesis: there is independency of these 2 variables, so there is no hospital with major numbers of C-sections.

Weight and length means compared to population values

According to the Stanford Medicine Children’s Health website (https://www.stanfordchildrens.org/en/topic/default?id=newborn-measurements-90-P02673 (https://www.stanfordchildrens.org/en/topic/default?id=newborn-measurements-90-P02673)), the population mean values are:
* Mean weight: 3.200kg
* Mean length: 50 cm

```
mean(Peso);mean(Lunghezza)
```

```
## [1] 3284.184
```

```
## [1] 494.6958
```

To test if weight and length values of our sample are significantly different from the population ones, we perform a t.test.

Weight

```
t.test(Peso, mu=3200, conf.level=0.95, alternative = "two.sided")
```

```
##
##  One Sample t-test
##
## data:  Peso
## t = 8.0108, df = 2497, p-value = 1.731e-15
## alternative hypothesis: true mean is not equal to 3200
## 95 percent confidence interval:
##  3263.577 3304.791
## sample estimates:
## mean of x
##  3284.184
```

Even if population mean (3200 g) and sample mean (3284g) are close values, the t-test suggests that these data are significantly different (p-value very close to 0, the H0 hypothesis of equality is rejected).

Length

```
t.test(Lunghezza, mu=500, conf.level=0.95, alternative = "two.sided")
```

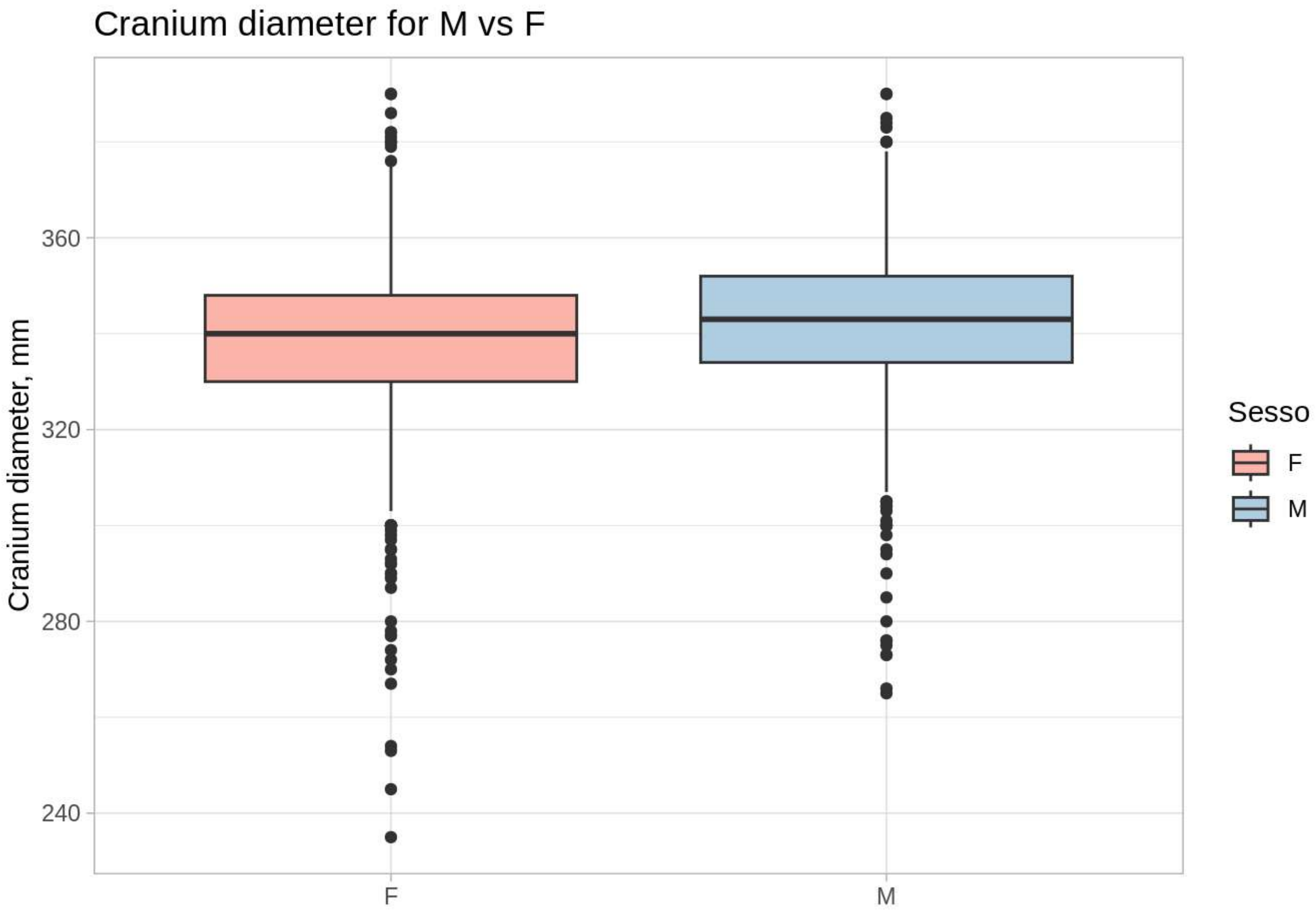
```
##
##  One Sample t-test
##
## data:  Lunghezza
## t = -10.069, df = 2497, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 500
## 95 percent confidence interval:
##  493.6628 495.7287
## sample estimates:
## mean of x
##  494.6958
```

In the same way, the t-test here establishs a relevant difference between population mean (500 cm) and sample mean (495 cm), with a significativity level of 5%.

Significative difference of anthropometric data between male and female

Cranium diameter

```
library(RColorBrewer)
ggplot(new_data)+
  geom_boxplot(aes(x = Sesso, y = Cranio, fill = Sesso))+
  labs(title = "Cranium diameter for M vs F",
       x = " ",
       y = "Cranium diameter, mm")+
  theme_light()+
  scale_fill_brewer(palette="Pastel1")
```



A t-test to compare group mean values is performed.

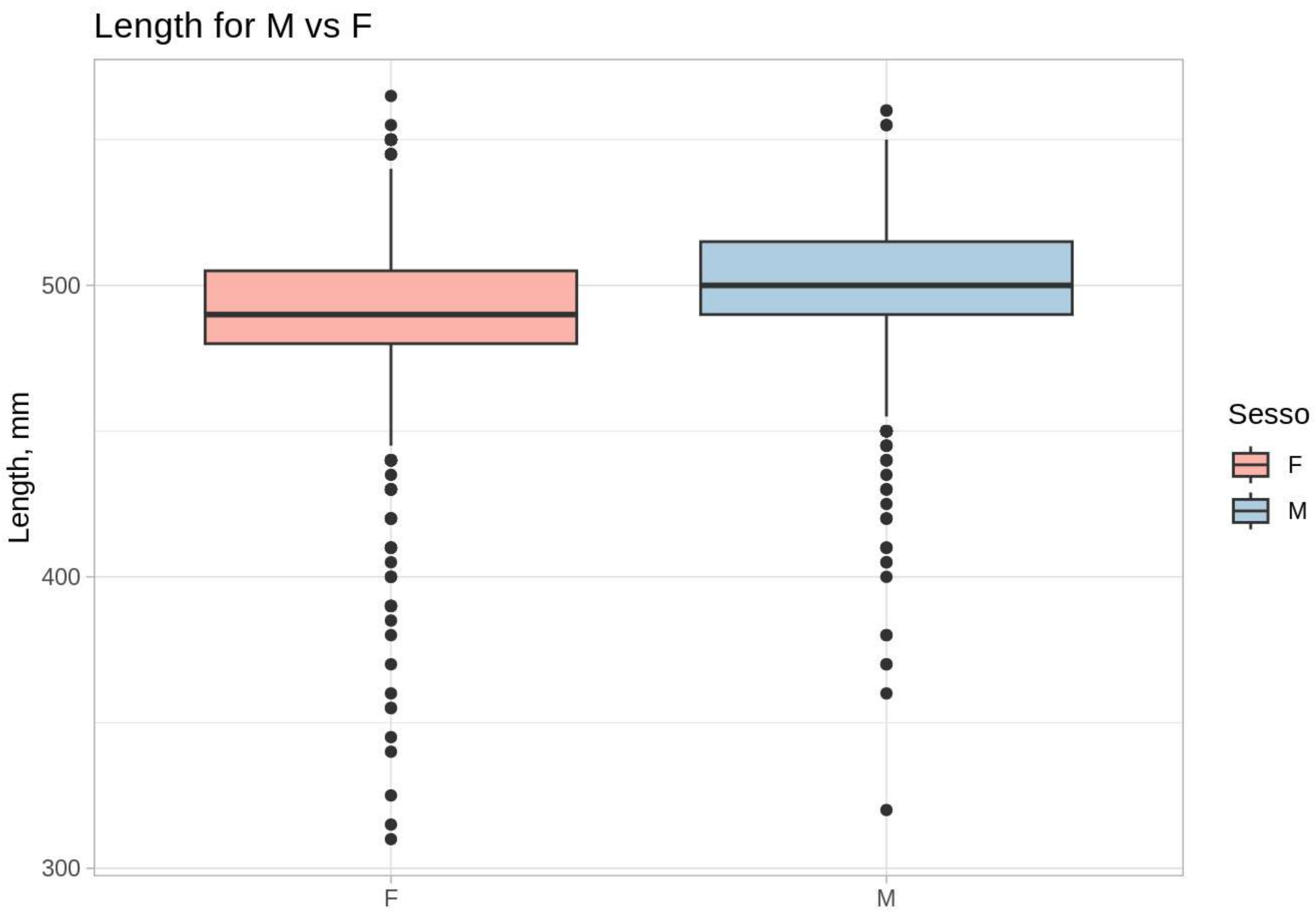
```
t.test(data = new_data, Cranio~Sesso, paired = F)
```

```
##
##  Welch Two Sample t-test
##
## data:  Cranio by Sesso
## t = -7.4366, df = 2489.4, p-value = 1.414e-13
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
##  -6.110504 -3.560417
## sample estimates:
## mean in group F mean in group M
##        337.6231        342.4586
```

The p-value is very low, so H0 hypothesis is rejected: there is relevant difference of cranium data between male and female.

Length

```
ggplot(new_data)+
  geom_boxplot(aes(x = Sesso, y = Lunghezza, fill = Sesso))+
  labs(title = "Length for M vs F",
       x = " ",
       y = "Length, mm")+
  theme_light()+
  scale_fill_brewer(palette="Pastel1")
```



A t-test to compare group mean values is performed.

```
t.test(data = new_data, Lunghezza~Sesso, paired = F)
```

```
##
##  Welch Two Sample t-test
##
## data:  Lunghezza by Sesso
## t = -9.5823, df = 2457.3, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
##  -11.939001  -7.882672
## sample estimates:
## mean in group F mean in group M
##        489.7641        499.6750
```

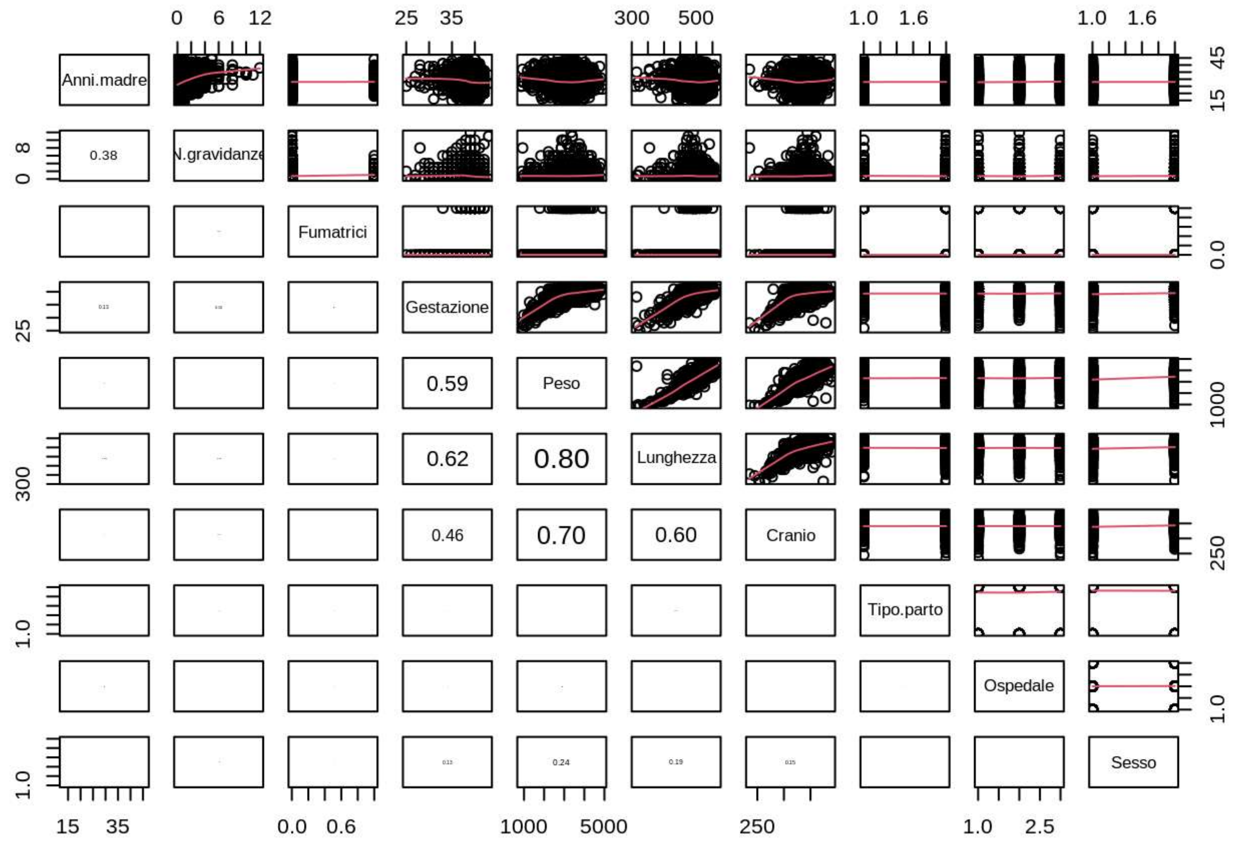
As a conclusion, it appears that both cranium diameter and length are significantly different in respect of the newborn sex.

Regression Model Creation

As preamble, we build a correlation matrix, together with the scatterplots of variables per pairs.

```
panel.cor <- function(x, y, digits = 2, prefix = "", cex.cor, ...)
{
    par(usr = c(0, 1, 0, 1))
    r <- abs(cor(x, y))
    txt <- format(c(r, 0.123456789), digits = digits)[1]
    txt <- paste0(prefix, txt)
    if(missing(cex.cor)) cex.cor <- 0.8/strwidth(txt)
    text(0.5, 0.5, txt, cex = cex.cor * r)
}

pairs(new_data, upper.panel = panel.smooth, lower.panel = panel.cor)
```



First, we observe the correlations between the response variable weight and the other quantitative variables:

Weight and mother age

```
cor(Peso, Anni.madre)
```

```
## [1] -0.02378138
```

As the correlation coefficient is very close to 0, we can say that, based on these sample data, there is no correlation between weight and mother age. Nevertheless, as age is a control variable, and is usual to be considered as very important about pregnancy, it will be kept in the regression model.

Weight and number of previous pregnancies

```
cor(Peso, N.gravidanze)
```

```
## [1] 0.002277118
```

With this very low value, it can be said that newborn's weight and number of previous pregnancies are not correlated variables.

Weight and gestation duration

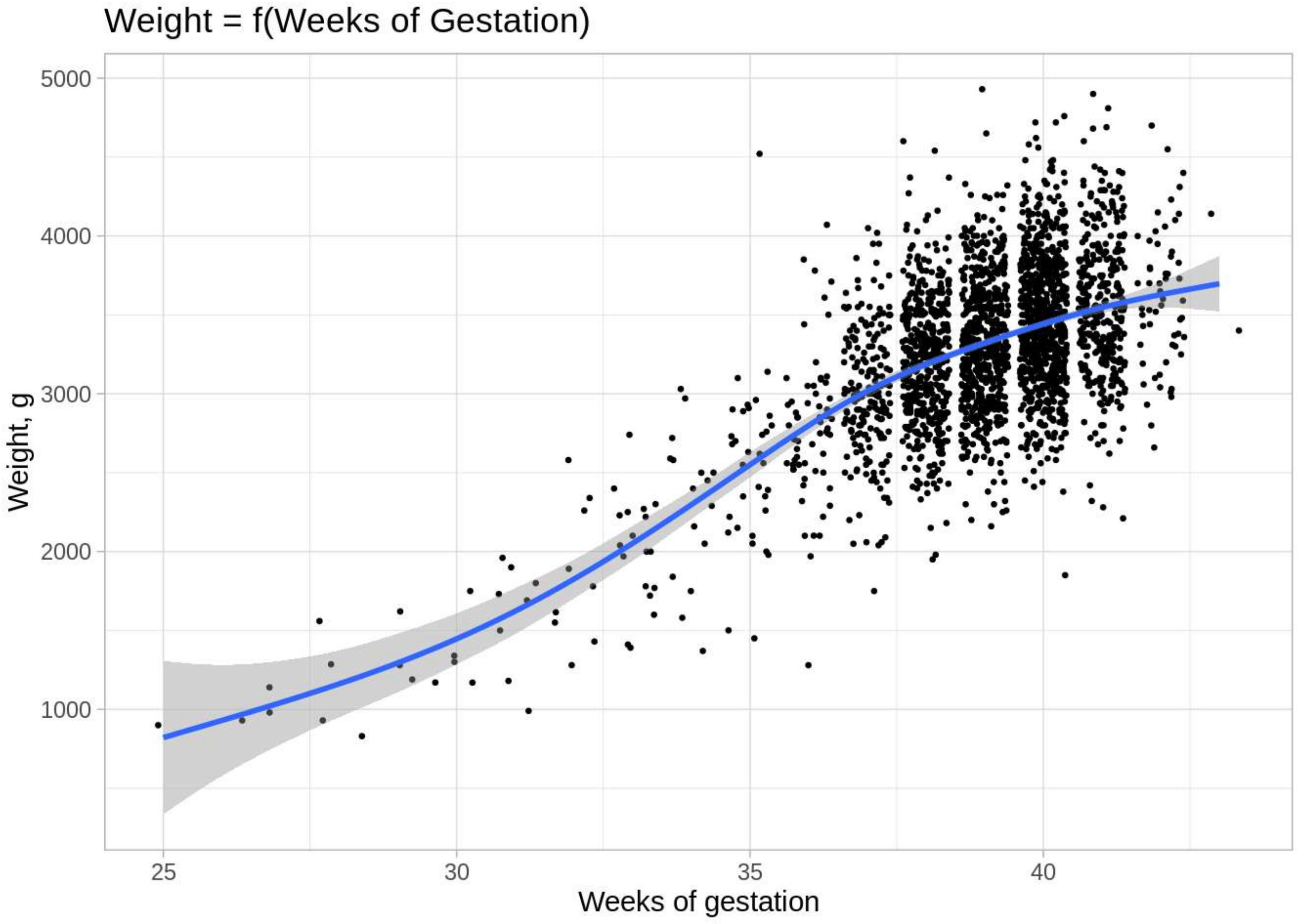
```
cor(Peso, Gestazione)
```

```
## [1] 0.5919592
```

With a correlation coefficient of nearly 0.6, it seems clear that these 2 variables are correlated: the newborn weight is higher if it stays longer in his/her mother's womb.

A plot is done to visualize this interaction.

```
ggplot(new_data)+
  geom_point(aes(x=Gestazione, y=Peso), position = "jitter", size = 0.5)+
  geom_smooth(aes(x=Gestazione, y=Peso))+
  labs(title= "Weight = f(Weeks of Gestation)",
       x = "Weeks of gestation",
       y = "Weight, g")+
  theme_light()
```



We observe a slighlty non-linear trend here.

Weight and anthropomorphic measurements

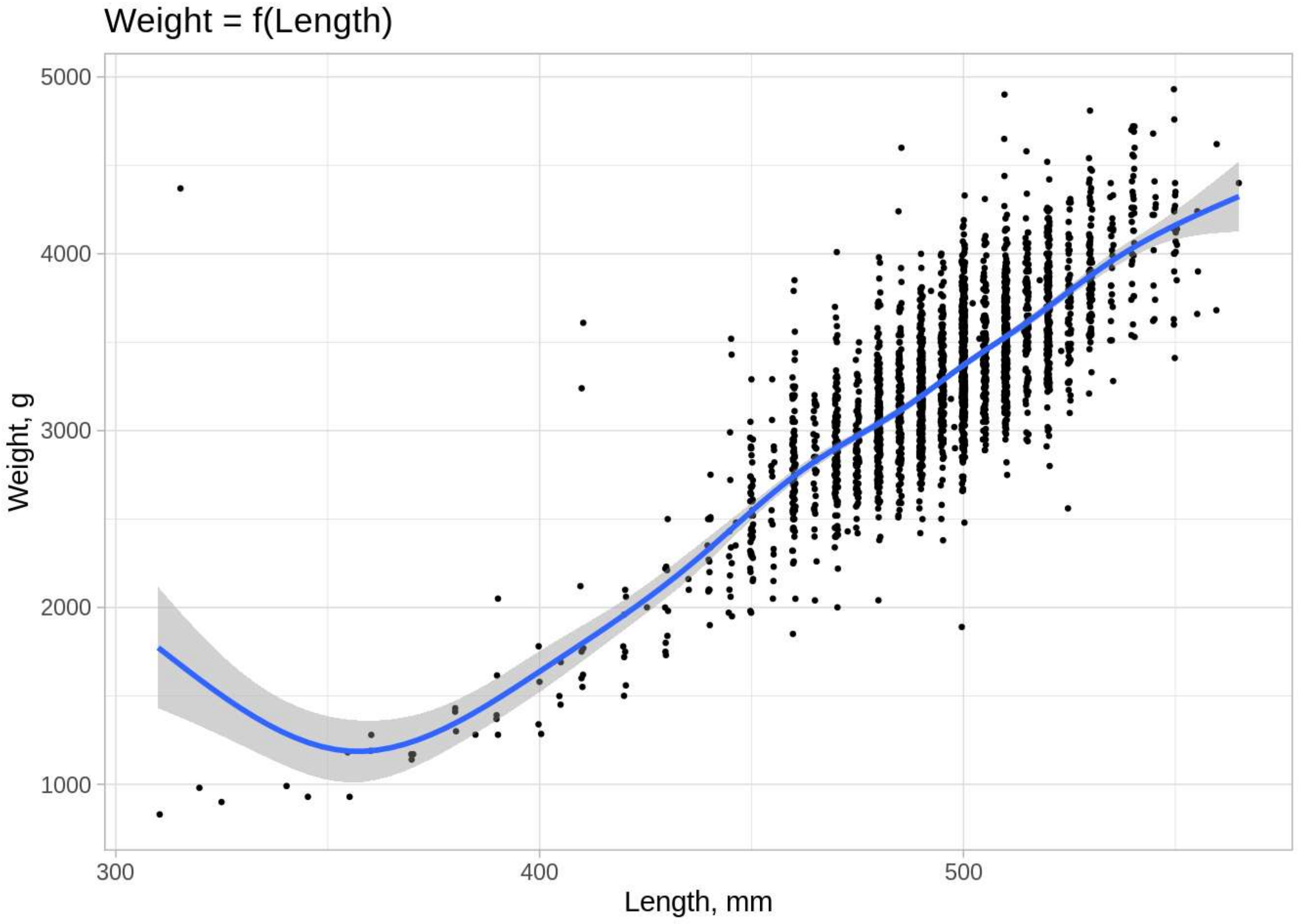
Weight and length

```
cor(Peso, Lunghezza)
```

```
## [1] 0.7960415
```

As logically expected, the weight and length variables are highly correlated. Let's plot this relation to investigate if it is linear or not.

```
ggplot(new_data)+
  geom_point(aes(x=Lunghezza, y=Peso), size = 0.5, position = "jitter")+
  geom_smooth(aes(x=Lunghezza, y=Peso))+
  labs(title= "Weight = f(Length)",
       x = "Length, mm",
       y = "Weight, g")+
  theme_light()
```



From around Length = 370 mm, the relation between weight and length seems to be linear. A point draws our attention:

```
which(Lunghezza == 315) ; Peso[1549]
```

```
## [1] 1549
```

```
## [1] 4370
```

It would corresponds to a newborn with atypical features: small length of 315 mm and important weight of 4.37 kg.

So far we keep this data, a further analysis with Cook distance will determine if it corresponds to an abnormal value.

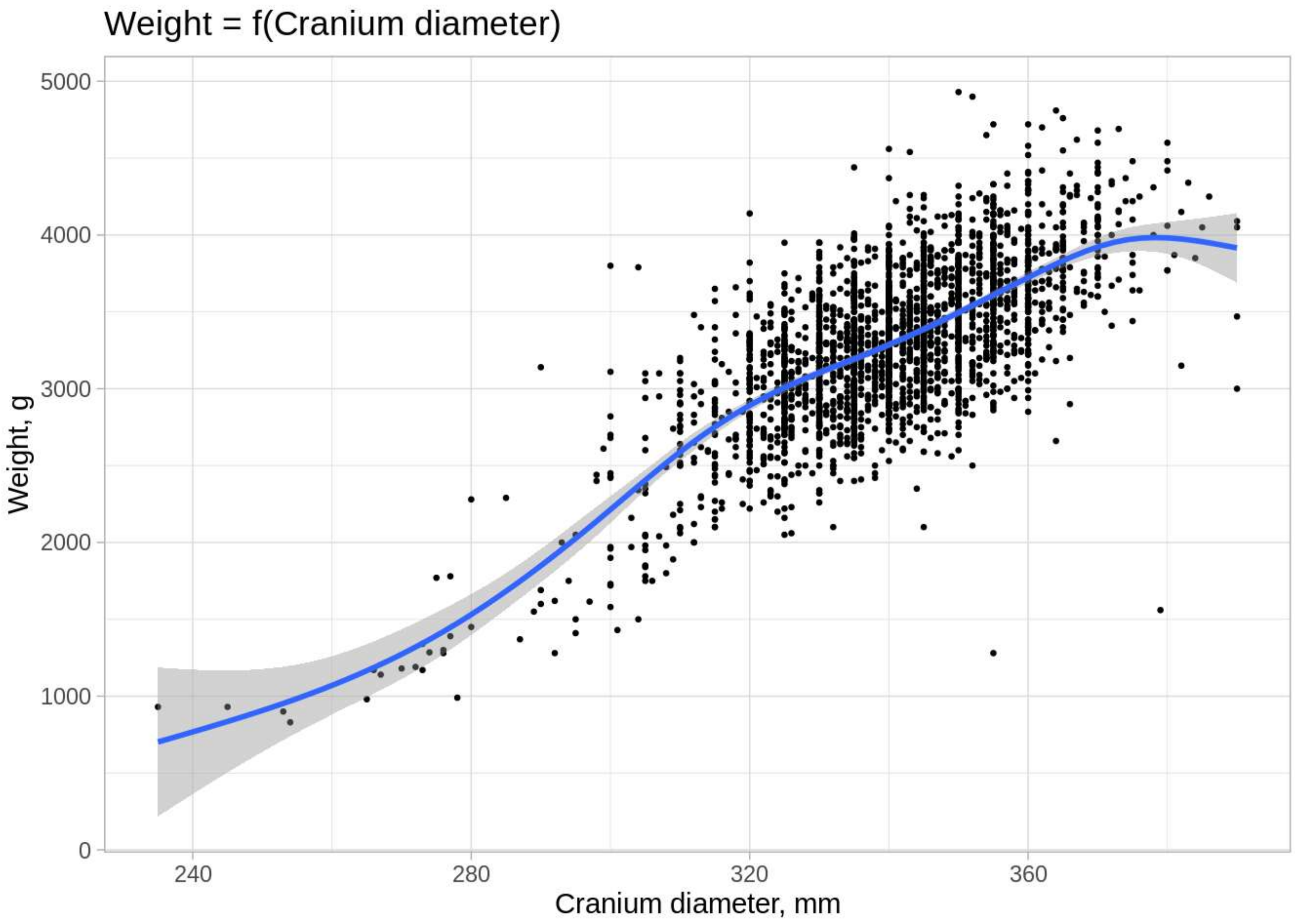
Weight and cranium diameter

```
cor(Peso, Cranio)
```

```
## [1] 0.7048438
```

As for length, there is a strong correlation between weight and cranium diameter.

```
ggplot(new_data)+
  geom_point(aes(x=Cranio, y=Peso), size = 0.5)+
  geom_smooth(aes(x=Cranio, y=Peso))+
  labs(title= "Weight = f(Cranium diameter)",
       x = "Cranium diameter, mm",
       y = "Weight, g")+
  theme_light()
```

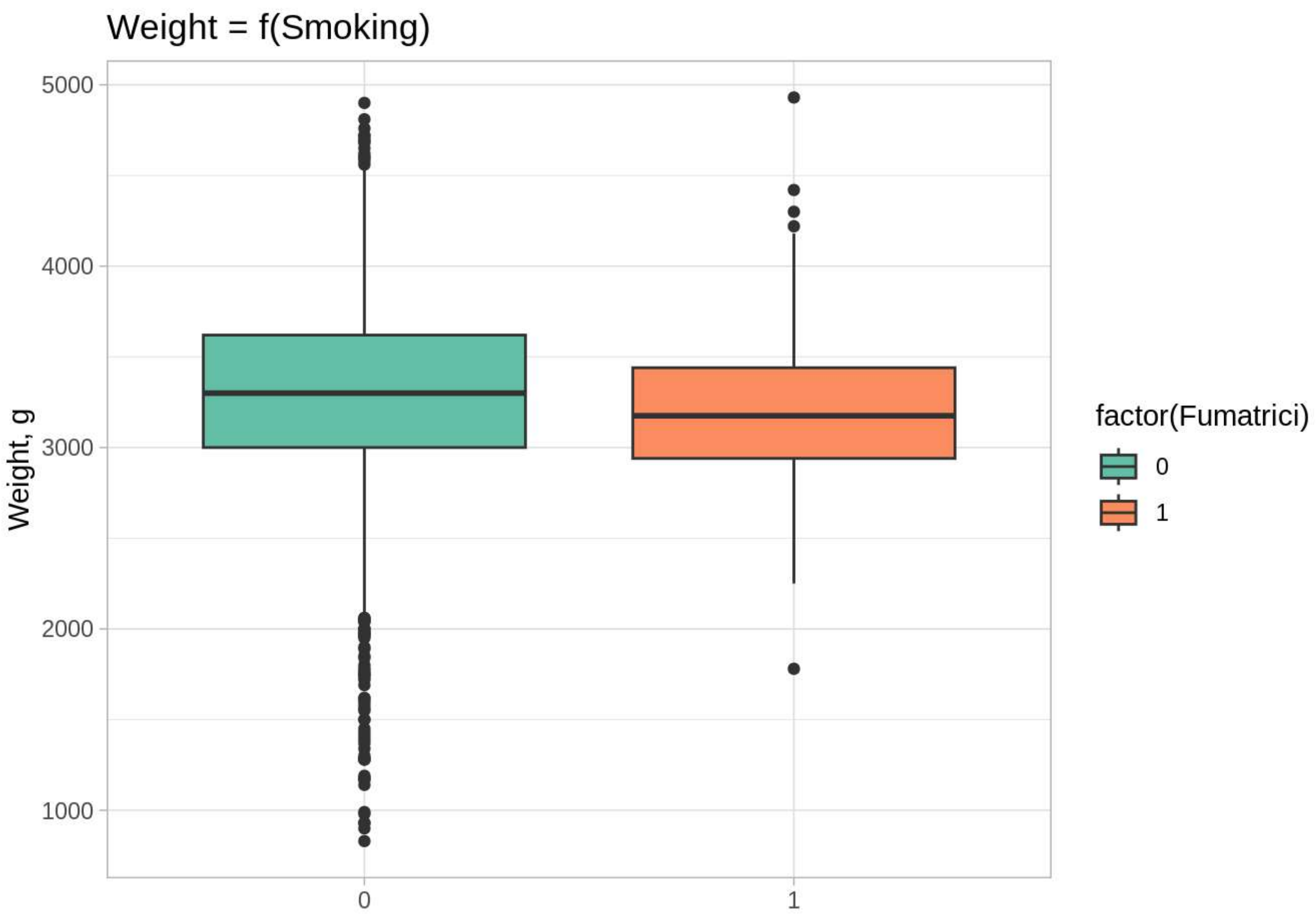


The relation doesn't seem to be perfectly linear, next, some quadratic effect can be investigated.

Then, we study correlations between the response variable "Peso" and the qualitative variables.

Smoking mother behaviour

```
ggplot(new_data)+
  geom_boxplot(aes(x = factor(Fumatrici), y = Peso, fill = factor(Fumatrici)))+
  labs(title = "Weight = f(Smoking)",
       x = " ",
       y = "Weight, g")+
  theme_light()+
  scale_fill_brewer(palette="Set2")
```



A t-test is done in order to test the correlation between these 2 variables.

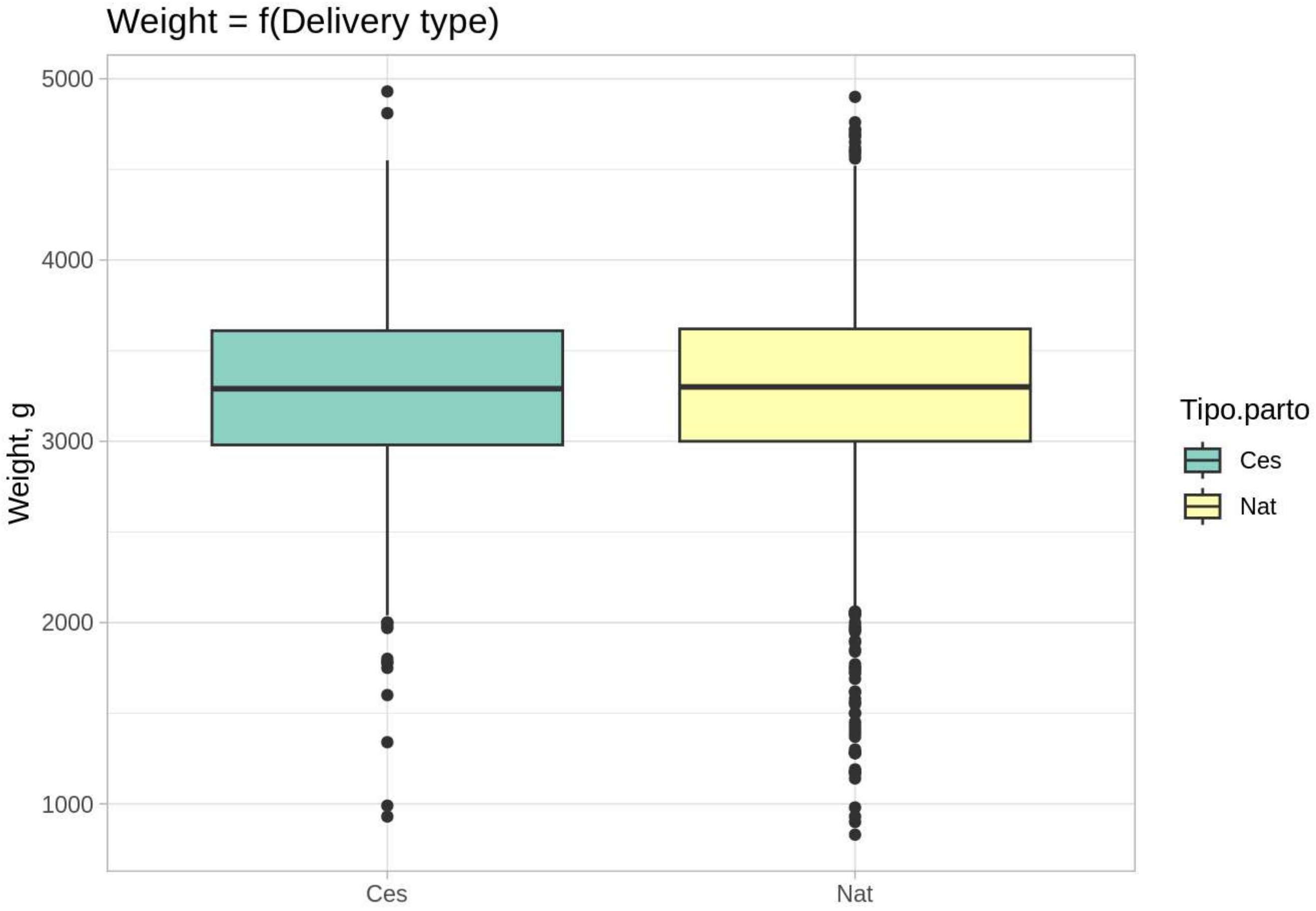
```
t.test(Peso~Fumatrici)
```

```
##
##  Welch Two Sample t-test
##
## data:  Peso by Fumatrici
## t = 1.0362, df = 114.12, p-value = 0.3023
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -45.5076 145.3399
## sample estimates:
## mean in group 0 mean in group 1
##        3286.262        3236.346
```

p-value equals to 30%: it is not possible to reject H0, so we don't conclude about a relevant difference of weight mean in respect of the smoking mother behaviour.

Delivery type

```
ggplot(new_data)+
  geom_boxplot(aes(x = Tipo.parto, y = Peso, fill = Tipo.parto))+
  labs(title = "Weight = f(Delivery type)",
       x = " ",
       y = "Weight, g")+
  theme_light()+
  scale_fill_brewer(palette="Set3")
```



A t-test is done in order to test the correlation between these 2 variables.

```
t.test(Peso~Tipo.parto)
```

```
##
##  Welch Two Sample t-test
##
## data:  Peso by Tipo.parto
## t = -0.13626, df = 1494.4, p-value = 0.8916
## alternative hypothesis: true difference in means between group Ces and group Nat is not equal to 0
## 95 percent confidence interval:
##  -46.44246  40.40931
## sample estimates:
## mean in group Ces mean in group Nat
##          3282.047          3285.063
```

Weight means of the 2 groups divided by delivery type are very close: in fact as visible on the boxplot it doesn't seem to be any difference, and this result is confirmed by the t-test which, with p-value = 89%, absolutly accept the hypothesis of means equality.

Even if this result seem logical, one could think that anticipating the delivery with a C-section would have led to lower weight values for C-section delivery type.

Hospital

No correlation is expected between hospital and weight, but a t-test is done to confirm this hypothesis.

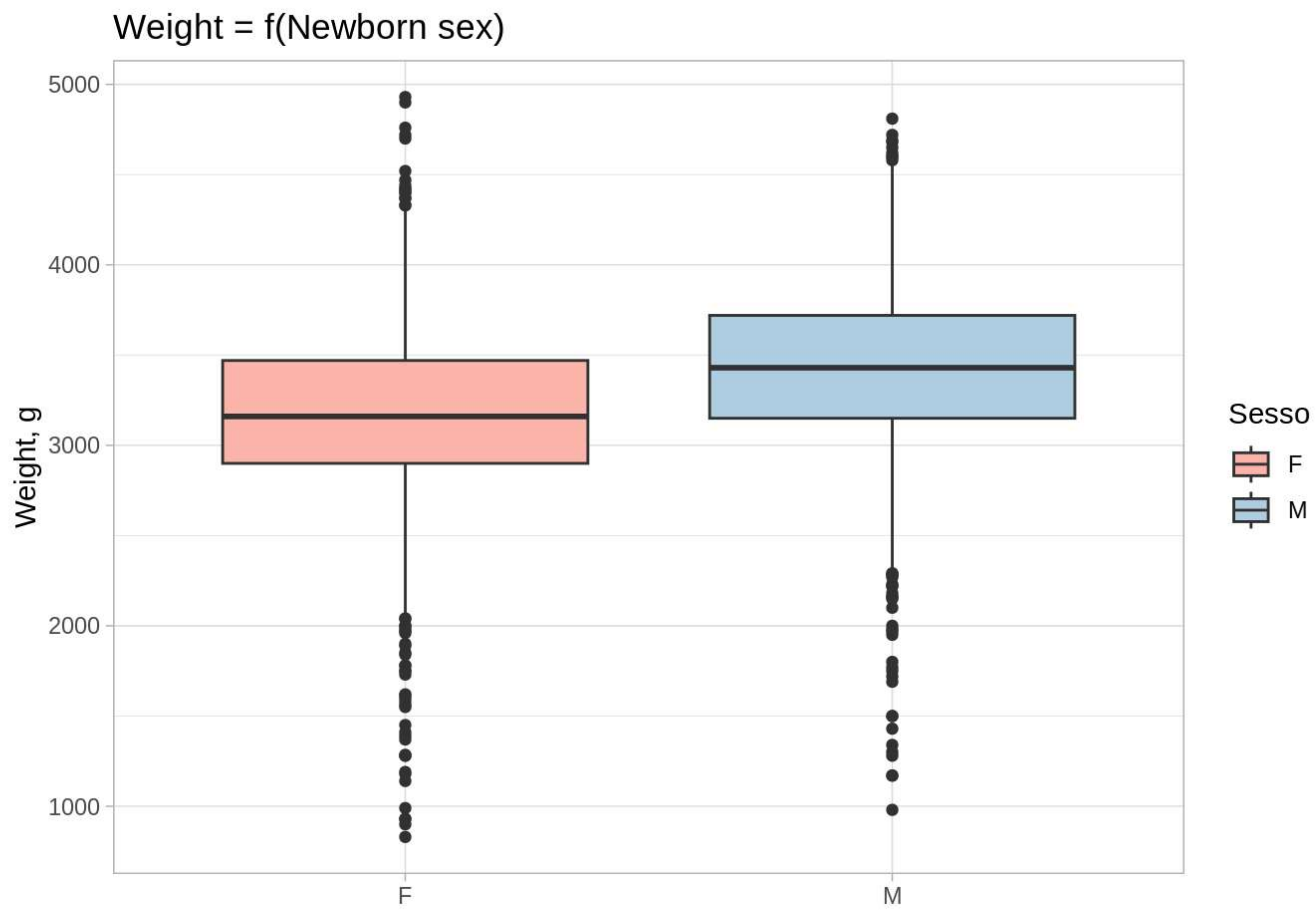
```
pairwise.t.test(Peso, Ospedale, paired = F, pool.sd = T, p.adjust.method = "bonferroni")
```

```
##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  Peso and Ospedale
##
##      osp1 osp2
## osp2 1.00 -
## osp3 0.33 0.32
##
## P value adjustment method: bonferroni
```

p-values are >30%, we accept the hypothesis of means equality: there is no correlation between weight value and the place of birth.

Newborn sex

```
ggplot(new_data)+
  geom_boxplot(aes(x = Sesso, y = Peso, fill = Sesso))+
  labs(title = "Weight = f(Newborn sex)",
       x = " ",
       y = "Weight, g")+
  theme_light()+
  scale_fill_brewer(palette="Pastel1")
```



A difference is visible between female weight and male weight. We test this hypothesis with a t-test.

```
t.test(Peso~Sesso)
```

```
##
##  Welch Two Sample t-test
##
## data:  Peso by Sesso
## t = -12.115, df = 2488.7, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
##  -287.4841 -207.3844
## sample estimates:
## mean in group F mean in group M
##        3161.061        3408.496
```

As p-value is close to 0, we conclude about the relevant difference of weight means between male and female.

Regression Model Creation

Model number 1

A first model including all variables is created.

```
mod1 <- lm(Peso~., data = new_data)
summary(mod1)
```

```
##
## Call:
## lm(formula = Peso ~ ., data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1123.26  -181.53   -14.45   161.05  2611.89
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -6735.7960   141.4790 -47.610  < 2e-16 ***
## Anni.madre        0.8018     1.1467   0.699   0.4845
## N.gravidanze     11.3812     4.6686   2.438   0.0148 *
## Fumatrici       -30.2741    27.5492  -1.099   0.2719
## Gestazione       32.5773     3.8208   8.526  < 2e-16 ***
## Lunghezza        10.2922     0.3009  34.207  < 2e-16 ***
## Cranio           10.4722     0.4263  24.567  < 2e-16 ***
## Tipo.partoNat    29.6335    12.0905   2.451   0.0143 *
## Ospedaleosp2    -11.0912    13.4471  -0.825   0.4096
## Ospedaleosp3     28.2495    13.5054   2.092   0.0366 *
## SessoM           77.5723    11.1865   6.934 5.18e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 274 on 2487 degrees of freedom
## Multiple R-squared:  0.7289, Adjusted R-squared:  0.7278
## F-statistic: 668.7 on 10 and 2487 DF,  p-value: < 2.2e-16
```

This model explains nearly 73% of the variability (adjusted R-squared).

As seen before, mother age, number of previous pregnancies, smoking habits, hospital are not correlated to the weight.

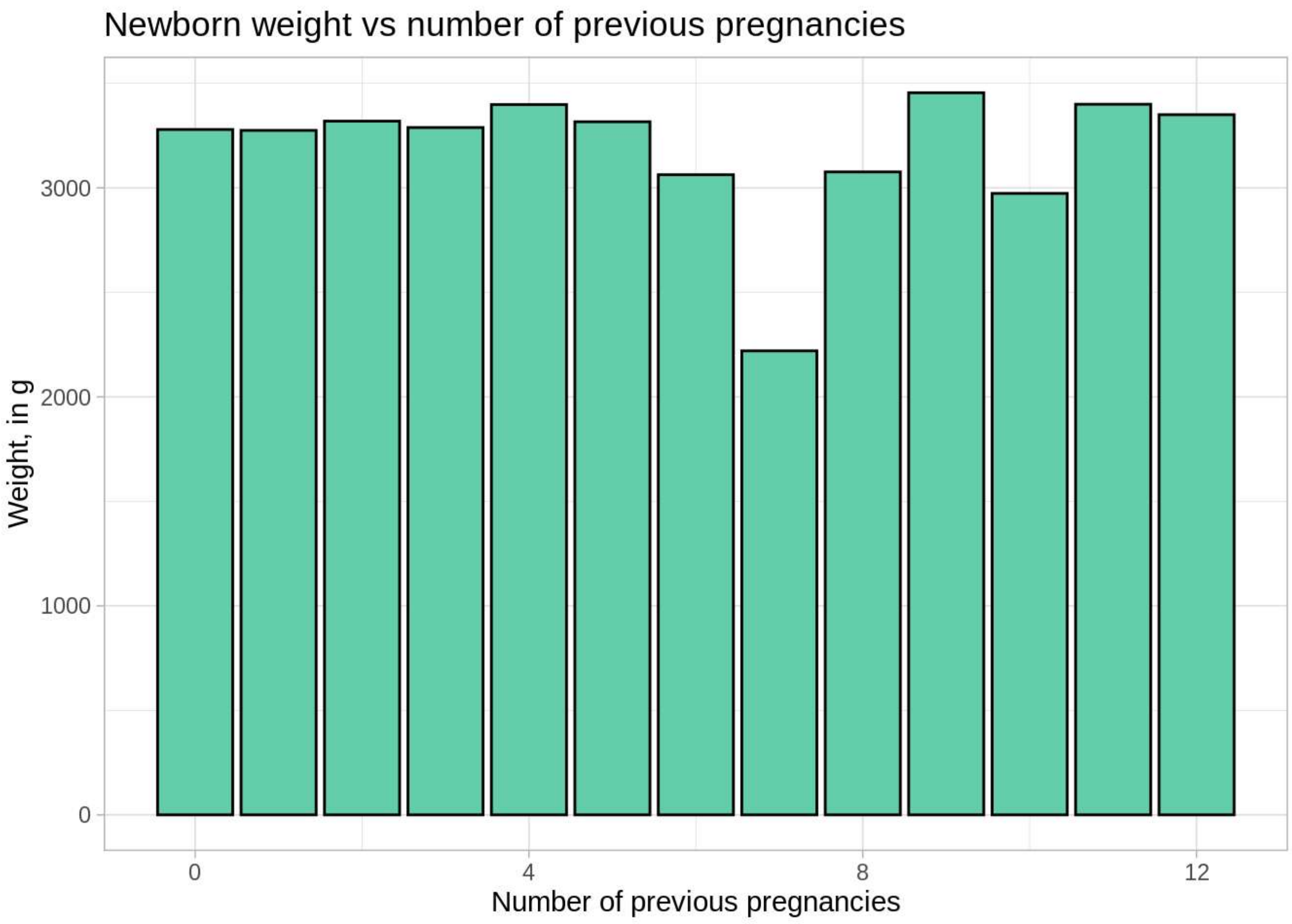
In detail:

- Mother age: p-value equals to 0.45, non significative effect. We decide to keep this variable as control variable ;
- Number of previous pregnancies: p-value of around 1.5%, so can be considered as significative. Beta coefficient equals to 11.4: it means that, all other things being equal, for each previous pregnancy of the mother, the newborn weight increases of 11 g.

To visualize this interaction, hereafter the corresponding plot.

```
library(dplyr)
by_Preg <-new_data %>%
  group_by(N.gravidanze)%>%
  summarise(media = mean(Peso))

ggplot(by_Preg, aes(x=N.gravidanze, y=media))+
geom_col(fill="mediumaquamarine", col="black")+
labs(title="Newborn weight vs number of previous pregnancies",
     x="Number of previous pregnancies",
     y="Weight, in g")+
theme_light()
```



The interaction is not obvious. The number of previous pregnancies variable is kept in the model for the moment. We will later study its impact on the quality linear model.

- Smoking mother habit: p-value is equals to 27%, as seen before it is not correlated with newborn weight value;
- Number of weeks of gestation: with a p-value close to 0, the result is very significative. The beta coefficient is about 32.6, it means that for each week more of gestation, the newborn gains 32,6g.
- Anthropomorphic data: both cranium diameter and length present a p-value close to 0, so a very relevant piece of information. Both have beta coefficients close to 10: it means that for each mm more of cranium diameter or length corresponds an increase of 10 g for the newborn.
- Delivery type: the p-value here is 1.4%, demonstrating the relevance of the variable data in the model. The beta coefficient of 29.6 here for the “Natural” modality of the variable means that meanly, the newborn weight is close to 30g more for the natural kind of delivery. It could be explained by the fact that in case of complications happening before the expected gestation end of 40 weeks, the newborn is delivered by C-section. So probably main part of the premature newborns experienced C-section.
- Newborn sex: with a very low p-value, sex is a relevant variable here. The beta coefficient of nearly 78 indicates that meanly, the male newborns weight 78 g more than females.

Model number 2

In the light of this information, a new model is created without the less relevant variables. Basically, we remove smoking mother habit and hospital.

```
mod2 <- lm(Peso~Anni.madre+N.gravidanze+Gestazione+Lunghezza+Cranio+Tipo.parto+Sesso, data = new_data)
summary(mod2)
```

```
##
## Call:
## lm(formula = Peso ~ Anni.madre + N.gravidanze + Gestazione +
##     Lunghezza + Cranio + Tipo.parto + Sesso, data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1139.50  -181.60   -14.59   160.14  2633.16
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -6737.9269   141.5747 -47.593  < 2e-16 ***
## Anni.madre        0.8793     1.1479   0.766   0.4438
## N.gravidanze     11.4176     4.6676   2.446   0.0145 *
## Gestazione       32.6300     3.8180   8.546  < 2e-16 ***
## Lunghezza        10.2839     0.3009  34.176  < 2e-16 ***
## Cranio           10.4896     0.4268  24.574  < 2e-16 ***
## Tipo.partoNat    30.1222    12.1038   2.489   0.0129 *
## SessoM           77.8374    11.2008   6.949 4.67e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 274.4 on 2490 degrees of freedom
## Multiple R-squared:  0.7278, Adjusted R-squared:  0.727
## F-statistic: 950.9 on 7 and 2490 DF,  p-value: < 2.2e-16
```

The beta coefficients didn’t change too much, and the adjusted R.squared is the same.

Model number 3

As seen before, we can also try to remove the number of previous pregnancies, and observe what happens then to the model.

```
mod3 <- update(mod2, ~. - N.gravidanze)
summary(mod3)
```

```
##
## Call:
## lm(formula = Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio +
##     Tipo.parto + Sesso, data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1143.29  -184.07   -15.52   161.15  2617.47
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -6748.7308   141.6472 -47.645  < 2e-16 ***
## Anni.madre        1.9144     1.0681   1.792   0.0732 .
## Gestazione       32.1537     3.8168   8.424  < 2e-16 ***
## Lunghezza        10.2496     0.3009  34.065  < 2e-16 ***
## Cranio           10.5733     0.4259  24.826  < 2e-16 ***
## Tipo.partoNat    29.3644    12.1120   2.424   0.0154 *
## SessoM           78.6331    11.2073   7.016 2.93e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 274.7 on 2491 degrees of freedom
## Multiple R-squared:  0.7271, Adjusted R-squared:  0.7264
## F-statistic:  1106 on 6 and 2491 DF,  p-value: < 2.2e-16
```

As from mod1 to mod2, here coefficients stay unchanged, as well as adjusted R.squared. We observe that the variable mother age became more relevantly correlated with weight, as its p-value turned 7%.

Model number 4

Now we consider quadratic effects, in particular for variable where the scatterplot shows non linear trend as number of weeks of gestation.

```
mod4 <- update(mod3, ~. +I(Gestazione^2))
summary(mod4)
```

```
##
## Call:
## lm(formula = Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio +
##     Tipo.parto + Sesso + I(Gestazione^2), data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1139.24  -184.38   -14.96   163.82  2639.80
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -4722.3507   899.1068  -5.252 1.63e-07 ***
## Anni.madre          1.9947     1.0678   1.868   0.0619 .
## Gestazione        -81.0806    49.7621  -1.629   0.1034
## Lunghezza          10.3537     0.3041  34.051  < 2e-16 ***
## Cranio             10.6684     0.4276  24.951  < 2e-16 ***
## Tipo.partoNat      28.8881    12.1035   2.387   0.0171 *
## SessoM             76.4063    11.2403   6.798 1.33e-11 ***
## I(Gestazione^2)     1.5121     0.6626   2.282   0.0226 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 274.5 on 2490 degrees of freedom
## Multiple R-squared:  0.7277, Adjusted R-squared:  0.7269
## F-statistic: 950.5 on 7 and 2490 DF,  p-value: < 2.2e-16
```

We observe that the effect of gestation at 1st grade becomes insignificant (p-value is up to 10%), while the quadratic effect of the variable has p-value below 5%. Considering the adjusted R-squared which is unchanged, to keep the model as simple as possible mod3 is prefered to mod4.

Model number 5

The cranium diameter variable shows a non linear trend vs weight too. We create mod5 with Cranio variable squared, from mod3.

```
mod5 <- update(mod3, ~. +I(Cranio^2))
summary(mod5)
```

```
##
## Call:
## lm(formula = Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio +
##     Tipo.parto + Sesso + I(Cranio^2), data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1133.43  -180.35   -13.76   163.11  2601.06
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -91.60711 1153.52855  -0.079   0.9367
## Anni.madre       1.90060    1.06118   1.791   0.0734 .
## Gestazione      38.54137    3.94788   9.763  < 2e-16 ***
## Lunghezza       10.48434    0.30164  34.758  < 2e-16 ***
## Cranio         -31.05845    7.17245  -4.330 1.55e-05 ***
## Tipo.partoNat   27.59571   12.03683   2.293   0.0220 *
## SessoM          73.86772   11.16433   6.616 4.49e-11 ***
## I(Cranio^2)      0.06159    0.01059   5.815 6.86e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 272.9 on 2490 degrees of freedom
## Multiple R-squared:  0.7308, Adjusted R-squared:   0.73
## F-statistic: 965.5 on 7 and 2490 DF,  p-value: < 2.2e-16
```

Both “Cranio” and “Cranio”^2 are significative, as they p-values are very low, but beta coefficient of “Cranio”^2 is close to 0, so the interaction with weight is not important.

To test the R-squared improvement between mod3 and mod5, a ANOVA test is performed.

```
anova(mod5, mod3)
```

```
## Analysis of Variance Table
##
## Model 1: Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio + Tipo.parto +
##     Sesso + I(Cranio^2)
## Model 2: Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio + Tipo.parto +
##     Sesso
##   Res.Df       RSS Df Sum of Sq      F    Pr(>F)
## 1   2490 185462712
## 2   2491 187980887 -1  -2518174 33.809 6.859e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

As the p-value is close to 0, the improvement of R-squared is considered as relevant. So far, mod5 is the best model.

Model number 6

Interactions between variables are studied. We first consider an interaction between mother age and gestation: it could be possible that oldest mothers experiment more often premature births.

```
mod6 <- update(mod5, ~. +Anni.madre*Gestazione)
summary(mod6)
```

```
##
## Call:
## lm(formula = Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio +
##     Tipo.parto + Sesso + I(Cranio^2) + Anni.madre:Gestazione,
##     data = new_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1117.3  -178.7   -14.0   163.7  2599.3
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          2165.18890 1352.17545   1.601  0.10945
## Anni.madre            -64.54520   20.90021  -3.088  0.00204 **
## Gestazione            -10.51294   15.90585  -0.661  0.50871
## Lunghezza              10.45375    0.30124  34.703  < 2e-16 ***
## Cranio                -33.04616    7.18651  -4.598 4.47e-06 ***
## Tipo.partoNat          26.12449   12.02370   2.173  0.02989 *
## SessoM                 74.00534   11.14399   6.641 3.82e-11 ***
## I(Cranio^2)             0.06453    0.01061   6.080 1.39e-09 ***
## Anni.madre:Gestazione    1.71255    0.53798   3.183  0.00147 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 272.4 on 2489 degrees of freedom
## Multiple R-squared:  0.7319, Adjusted R-squared:  0.731
## F-statistic: 849.1 on 8 and 2489 DF,  p-value: < 2.2e-16
```

Weeks of gestation as single effect is not significative anymore. The added interaction is real (p-value = 0.5% and beta coefficient of 1.7). Let's evaluated the relevance of the adjusted R-squared for mod6 vs mod5.

```
anova(mod6, mod5)
```

```
## Analysis of Variance Table
##
## Model 1: Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio + Tipo.parto +
##     Sesso + I(Cranio^2) + Anni.madre:Gestazione
## Model 2: Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio + Tipo.parto +
##     Sesso + I(Cranio^2)
##   Res.Df       RSS Df Sum of Sq      F   Pr(>F)
## 1   2489 184710712
## 2   2490 185462712 -1   -752000 10.133 0.001474 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

p-value of this test is 0.15%, which means that this model is significantly better.

Model number 7

Interaction between type of delivery and gestation duration is studied in mod7, from mod6.

```
mod7 <- update(mod6, ~. +Tipo.parto*Gestazione)
summary(mod7)
```

```
##
## Call:
## lm(formula = Peso ~ Anni.madre + Gestazione + Lunghezza + Cranio +
##     Tipo.parto + Sesso + I(Cranio^2) + Anni.madre:Gestazione +
##     Gestazione:Tipo.parto, data = new_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1124.17  -181.05   -12.26   163.19  2600.76
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)             2659.76461 1382.75064   1.924  0.05453 .
## Anni.madre               -66.16899   20.91425  -3.164  0.00158 **
## Gestazione               -20.01304   16.85727  -1.187  0.23526
## Lunghezza                 10.44301    0.30119  34.672  < 2e-16 ***
## Cranio                   -33.72876    7.19506  -4.688 2.91e-06 ***
## Tipo.partoNat           -420.35703  263.46958  -1.595  0.11073
## SessoM                    74.71627   11.14767   6.702 2.53e-11 ***
## I(Cranio^2)                0.06553    0.01063   6.167 8.10e-10 ***
## Anni.madre:Gestazione      1.75222    0.53829   3.255  0.00115 **
## Gestazione:Tipo.partoNat  11.44381    6.74598   1.696  0.08994 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 272.3 on 2488 degrees of freedom
## Multiple R-squared:  0.7322, Adjusted R-squared:  0.7312
## F-statistic: 755.7 on 9 and 2488 DF,  p-value: < 2.2e-16
```

The interaction variation between “Tipo.parto” and “Gestazione” is not relevant, with p-value of 9%.

Regression model mod6 appears to be the most relevant.

Optimal Model Selection

Akaike and Bayes information criteria are calculated for all the created models.

```
AIC(mod1, mod2, mod3, mod4, mod5, mod6, mod7)
```

```
##      df      AIC
## mod1 12 35145.57
## mod2  9 35150.08
## mod3  8 35154.07
## mod4  9 35150.85
## mod5  9 35122.38
## mod6 10 35114.23
## mod7 11 35113.35
```

According to AIC data, mod7 is the best, and then mod6 with the second lower value.

```
BIC(mod1, mod2, mod3, mod4, mod5, mod6, mod7)
```

```
##      df      BIC
## mod1 12 35215.45
## mod2  9 35202.49
## mod3  8 35200.66
## mod4  9 35203.26
## mod5  9 35174.79
## mod6 10 35172.47
## mod7 11 35177.40
```

Based on BIC values, mod6 is the best.

Model Quality Analysis

We consider mod6 as our final and best regression model.

RMSE calculation

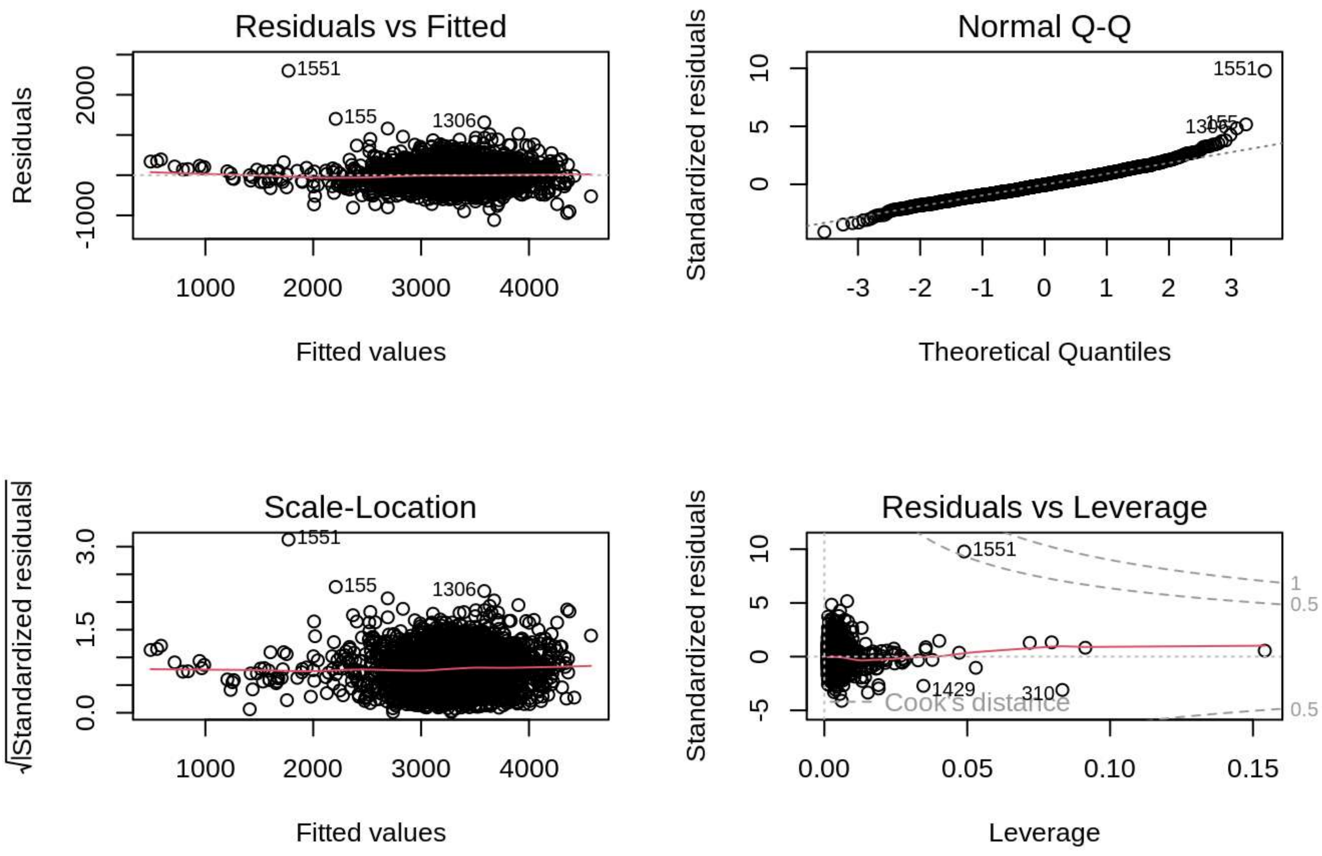
```
RMSE_6 <- sqrt(mean(mod6$residuals^2))
print(RMSE_6)
```

```
## [1] 271.9254
```

RMSE measures the average difference between values predicted by the model and the actual values. So using mod6, the average gap between predicted value and reel one is nearly 272g on the variable weight.

Residuals analysis

```
par(mfrow = c(2,2))
plot(mod6)
```



Residuals mean value

```
mean(residuals(mod6))
```

```
## [1] -1.376428e-14
```

As requested for linear regression model, residuals mean is equal to 0.

Residuals Gaussian distribution

```
library(lmtest)
shapiro.test(residuals(mod6))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(mod6)
## W = 0.97489, p-value < 2.2e-16
```

p-value is close to 0: residuals don't follow a normal distribution.

Looking at the Q-Q plot, extreme values indeed are out of the bisector plot.

Homoscedasticity

```
bptest(mod6)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  mod6
## BP = 91.261, df = 8, p-value = 2.579e-16
```

As p-value is very low, we reject H0, homoscedasticity hypothesis, even if on the plot, the points seem to be casually placed around an horizontal value.

Residuals non-correlation

```
dwtest(mod6)
```

```
##
##  Durbin-Watson test
##
## data:  mod6
## DW = 1.9506, p-value = 0.1083
## alternative hypothesis: true autocorrelation is greater than 0
```

As p-value is equals to 10%, we don't reject H0: residuals are not correlated.

Cook distance analysis

Leverage and outliers are identified using Cook distance, which plot shows only the 1551 point on the 0.5 line.

We decide to keep this value, as far from the 1 "alert" line. Even if it corresponds to strange values, as we don't have the certainty if it's an error, it's better to keep it.

Prediction

Weight of a female newborn, from a mother with 2 previous pregnancies, and a gestation of 39 weeks.

```
predict(mod6,
        newdata = data.frame(
  Sesso="F",
  Gestazione=39,
  N.gravidanze=2,
  Anni.madre=mean(Anni.madre),
  Lunghezza=mean(Lunghezza),
  Cranio=mean(Cranio),
  Tipo.parto="Nat"))
```

```
##        1
## 3240.584
```

The predicted weight is equal to 3241 g.

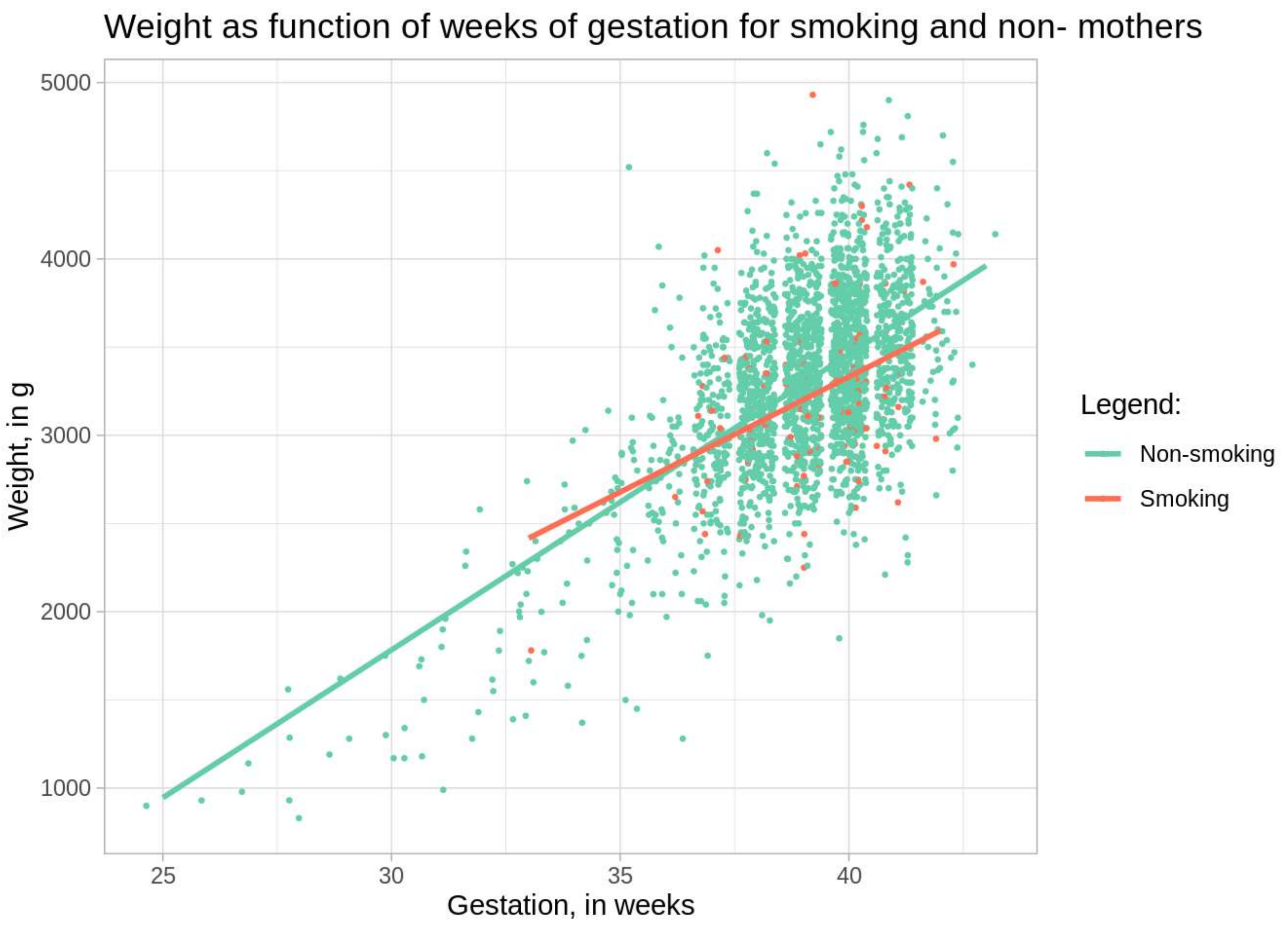
Visualization

Weight as function of weeks of gestation and sex or smoking habits

```
ggplot(data=new_data)+
  geom_point(aes( x = Gestazione, y = Peso, col = Sesso), position = "jitter", size = 0.5)+
  geom_smooth(aes( x = Gestazione, y = Peso, col = Sesso), se = F, method = "lm")+
  labs(title = "Weight as function of weeks of gestation for male and female",
       x = "Gestation, in weeks",
       y= "Weight, in g")+
  theme_light()
```

Weight newborn increases with weeks of gestation. As male weight is higher as female's, the curve for male is a little higher than female's one, but the slope is the same for both.

```
ggplot(data=new_data)+
  geom_point(aes( x = Gestazione, y = Peso, col = factor(Fumatrici)), position = "jitter", size = 0.5)+
  geom_smooth(aes( x = Gestazione, y = Peso, col = factor(Fumatrici)), se = F, method = "lm")+
  labs(title = "Weight as function of weeks of gestation for smoking and non- mothers",
       x = "Gestation, in weeks",
       y= "Weight, in g")+
  theme_light()+
  scale_color_manual(name = "Legend: ", values=c("aquamarine3", "coral1"), labels = c("Non-smoking", "Smoking"))
```



There is a lower slope for the data about smoking mothers: for long gestations (beyond 36 weeks), weight newborn gets slighlty lower than newborns from non-smoking mothers.

Conclusion

A regression model has been developped in order to explain which variables can affect newborns weight and to predict it. In this study the most useful part is the prediction: it is important to be able to anticipate this data before birth, to organize human and material resources to take care about most fragile newborns, the ones with lower weights.

The most relevant variables are weeks of gestation and sex, and as control variables length and cranium diameter increase with the weight (the baby becomes bigger, so heavier and his measurements grow as well).

As surprising as it may sound, some variables as smoking mother habits don't influence the weight, and as singular effect the mother age is not relevant either.

In the model kept as the best one, interaction between mother age and gestation as been found as relevant: for older mothers, generally gestation is shorter. It corresponds to more risky pregnancies that often don't reach the normal term.

```
ggplot(data=new_data)+
  geom_point(aes( x = Anni.madre, y = Gestazione), position = "jitter", size = 0.5)+
  geom_smooth(aes( x = Anni.madre, y = Gestazione), se = F, method = "lm")+
  labs(title = "Gestation duration as function of mother age",
       x = "Mother age",
       y= "Gestation, in weeks")+
  theme_light()
```